Sharp Hall localization and exact curvature in graph complements

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Abstract

Let G be a connected simple graph whose complement has pair codegree at most k. We study equality in the triangle upper bound for zero-idleness Ollivier curvature. A negative-slope weighted Hall obstruction has numerator at most $k^2+k+2-n$, giving a sharp cutoff at $n=k^2+k+1$ for $k\geq 3$. At the boundary there are two numerical witness types, exact curvature deficit $1/(ab)$, and repair by one universal vertex. At codegree two and every order, an explicit local pattern classifies all defects and their pendant/nonpendant curvature values. Above the cutoff, every defective edge has $\mathcal{O}(k)$ exclusive targets. Two disjoint distance-three pairs would force $n\leq 4k$, so the large-order residual cost-three support is a star. A canonical maximum-slope Hall set, obtained from one ordinary matching, evaluates the full weighted deficit and its distance-three correction without constructing common-source rows. This gives $\mathcal{O}(k^3)$ work per large-order candidate after indexed preprocessing. An implemented all-orders hybrid uses $\mathcal{O}(k^5n^2)$ word-RAM operations; a classical small-vertex-cover flow substitution gives the theoretical bound $\mathcal{O}(k^4n^2)$. The counting, Hall, matching and flow machinery receive explicit prior-art credit. Exact checks and partial Lean formalization corroborate the arguments. Native transport wins the generic solver benchmarks, while structural reductions help selected dense all-edge workloads; cold indexing can lose badly for small query batches. Priority and independent human review remain unresolved.

1 Problem, result, and relation to previous work

For a vertex x of positive degree, let μ_x be the uniform probability measure on its open neighborhood $N_G(x)$. For an edge xy, define

$$\kappa_0(x,y)=1-W_1(\mu_x,\mu_y),$$

where transport cost is shortest-path distance in G. The subscript emphasizes zero idleness: neither closed-neighborhood measures nor the Lin–Lu–Yau limiting convention is used. Write $H=\overline{G}$ and assume

$$|N_H(u)\cap N_H(v)|\leq k \qquad (u\neq v). \tag{1}$$

This is equivalent to excluding $K_{2,k+1}$ as a not necessarily induced subgraph of H. It does not bound the maximum degree of H.

The structural results concern a marked edge xy oriented so that $a=d_G(x)\leq b=d_G(y)$. Write $d=b-a$, $c=|N_G(x)\cap N_G(y)|$, $\beta=|N_G(y)\setminus N_G(x)|$, and $b_k=\max\{k+1,2k-2\}$. The established triangle bound is $\kappa_0(x,y)\leq c/b$ [2].

*Proposed attribution in a research draft, subject to the named author's acceptance. Significant AI assistance and the current review status are disclosed in Section 10.

Theorem 1.1 (Structural localization, summary)**.** *Let $k \geq 2$. Every positive scaled Hall deficiency is at most*

$$U = (b_k - \beta)a + d(k-1).$$

Thus $U \leq 0$ implies $\kappa_0(x,y) = c/b$. Every edge surviving this scalar test satisfies $\min(a,\beta) < 2b_k$. A deficient Hall witness with negative coefficient of a has $n \leq k^2+k+1$. Above that order, every defective edge satisfies $\beta \leq b_k$. For $k \geq 3$, the order cutoff and the persistent bound $\beta \leq 2k-2$ are sharp. For $k = 2$ no negative-slope deficient witness exists.

The scalar test, proved in Proposition 2.2, applies at every order; it need not decide equality on every edge. The finite-order cutoff is proved in Theorem 3.1. Their combination is useful because the residual transport may have many common sources, and distance-three transfers can remain necessary even at arbitrarily large order.

Corollary 1.2 (Matching hybrid)**.** *For $k \geq 2$, under condition (1), all edge values can be computed exactly in $\mathcal{O}(k^5n^2)$ word operations by the implemented matching/dense-Dinic hybrid. For $n \geq k^2+k+2$, matching and shortcuts alone take $\mathcal{O}(k^3n^2)$ including preprocessing. Replacing only the small-order flow backend by the classical AOST specialized algorithm gives the theoretical all-orders bound $\mathcal{O}(k^4n^2)$, with $\mathcal{O}(kn^2)$ words of space and unreduced output. The latter backend is unimplemented. The proof is in Section 7; no optimal or best-known k exponent is claimed. The $k = 1$ triangle-equality case takes $\mathcal{O}(n^2)$.*

The direct network design below remains a separately verified baseline and fallback. The final matching reduction avoids its common-source rows at large order.

Corollary 1.3 (Direct network design)**.** *For $k \geq 2$, condition (1) can be recognized and, on a connected accepted graph, all edge curvatures can be computed exactly in*

$$\mathcal{O}\big(k^5 \log(k+1) n^2\big)$$

word-RAM operations, using $\mathcal{O}(kn^2)$ words of space and unreduced rational output. The algorithm described below uses ordinary current-arc Dinic in a classical primal-dual transport method. It uses dense transport when $n \leq k^2+k+1$ and a segment representation otherwise. For $k = 1$, all edges attain the triangle bound and $\mathcal{O}(n^2)$ time suffices. Explicitly listing all edge values requires quadratic time in n in the worst case.

The corresponding algorithm is implemented in the supplied Python code. The deterministic RAM bound assumes indexed adjacency and pair-witness tables and constant-time exact arithmetic on $\mathcal{O}(\log n)$-bit words, with k capped at $\max(1, n-2)$. Python sets and dictionaries instead provide expected-time access. Reduced `Fraction` outputs add normalization costs omitted from the unreduced-output theorem.

The closest prior framework is Chen, Liu, and You [9], who study positive Lin–Lu–Yau curvature under forbidden subgraphs in complements. Their Claim 5.2 already bounds the number of missing neighbors into the relevant exclusive neighborhood by k; their Section 5 also reserves an endpoint when counting common nonneighbors. We explicitly credit these local counting principles. The present sharp cutoff classifies transient defects in equality for zero-idleness curvature, rather than positivity of Lin–Lu–Yau curvature. For $k \geq 8$, our sharp cutoff construction has $n = k^2+k+1 \geq \max\{k^2+1, 8k+8\}$ and hence lies within their Theorem 1.2 with $t = k+1$: every edge has positive Lin–Lu–Yau curvature, while the marked edge still has a transient triangle defect. This is a distinct conclusion inside their positivity regime; it is not a claim that the complement viewpoint or its basic counting tools are new.

Kelly [8] proves integrality of graph potentials and develops reduced local linear programs. Hehl [10] gives optimal-assignment formulas for regular graphs, including zero-idleness curvature; in the equal-degree case, cancellation of the common neighborhood and an assignment formulation are therefore established. Matching-based curvature results also precede this work [11]. Closed-neighborhood computational results [12] must be compared with their measure convention intact.

Bonini et al. [1], Theorem 4.6 and its preceding alternating-path lemmas, already express condensed (LLY) curvature through maximum-matching deficiency in strongly regular graphs. Matching and alternating reachability are therefore established curvature tools, not new features in isolation.

Li and Lu [3], Corollaries 2–3, already characterize equality in an LLY upper bound by unequal-degree weighted Hall conditions. Relative to our scaled residual, their LLY problem removes target x of demand a and reduces source y from b to d; for each nonempty $S \subseteq B \setminus \{x\}$ its Hall numerator is $F(S) + a$, rather than $F(S)$. Weighted Hall and this two-weight pattern are established machinery. Asai and Yamada [4] directly study zero-idleness triangle equality for complements of matchings, stars under their size condition, and cycles C_m with $3 \le m \le n-1$. These special-family results are relevant predecessors, not the general cutoff claimed here.

The extremal constructions also need explicit attribution: the persistent example is exactly a member of the four-block family in Chen–Liu–You, Section 6 [9]; the finite-cutoff example modifies that template. The unmodified family already gives a transient zero-idleness defect at $n = k^2+k-1$, as calculated in the accompanying priority audit. The proposed sharp inequality and boundary analysis must be assessed against that close overlap.

Successive shortest-path min-cost flow, feasible potentials, blocking flows, unbalanced-network analysis [14], and identical-row aggregation are established machinery. The segment tree only shares ordinary cost-one arcs. The candidate contribution is the sharp localization, the universal scalar test and resulting small-side property, and their use with indexed transport construction. We claim neither a new general flow algorithm nor an optimal dependence on k. Capacities retain $\mathcal{O}(\log n)$ bits, so no polynomial kernel in total bit length is claimed. A bounded literature audit does not establish priority.

2 Residual transport and Hall obstructions

Orient an edge xy so that $a = d_G(x) \le b = d_G(y)$, and put

$$A = N_G(x) \setminus N_G(y), \qquad B = N_G(y) \setminus N_G(x),$$
$$C = N_G(x) \cap N_G(y), \qquad R = V(G) \setminus (N_G(x) \cup N_G(y)).$$

Let $\alpha = |A|$, $\beta = |B|$, $c = |C|$, $r = |R|$, and $d = b - a$. Then

$$y \in A, \quad x \in B, \quad a = \alpha + c, \quad \beta = \alpha + d, \quad n = a + \beta + r, \quad r \le k. \tag{2}$$

Also $d_H(x) = \beta - 1 + r$ and $d_H(y) = \alpha - 1 + r$.

An optimal metric transport plan can leave mass $1/b$ at each vertex of C. For completeness, maximize total diagonal mass among optimal plans. If a common vertex v has less diagonal mass than the smaller of its two marginal masses, there are positive transfers $u \to v$ and $v \to w$, with $u, w \ne v$. Replace an equal amount of those transfers by $u \to w$ and $v \to v$. The triangle inequality prevents a cost increase, while diagonal mass increases, a contradiction. Repeating this argument gives maximal common-mass cancellation.

The remaining sources are A, with supply $1/a$ per vertex, and C, with supply $d/(ab)$ per vertex; zero supplies may be omitted. The targets are B, each demanding $1/b$. The residual total is

$$T = \frac{\beta}{b} = 1 - \frac{c}{b}. \tag{3}$$

All source-target pairs are distinct and cost at least one. Consequently $\kappa_0(x, y) \le c/b$, with equality precisely when the residual supplies admit transport using only edges of G.

Lemma 2.1 (Hall witness)**.** *If $\kappa_0(x, y) < c/b$, there is a set $S \subseteq B \setminus \{x\}$ such that, writing $s = |S|$ and letting p, q count the sources in A, C adjacent to no member of S,*

$$2 \le s \le k, \qquad p + q \le k - 1, \qquad p + 1 \le \alpha, \qquad q \le c, \tag{4}$$

and

$$F = (s + p - \beta)a + d(p + q) > 0. \tag{5}$$

Conversely, any such set with $F > 0$ certifies failure of equality. If $p \ge 2$, then $s \le k - 1$.

Proof. Weighted Hall feasibility follows directly by putting source supplies on arcs from a super-source, target demands on arcs to a super-sink, and unrestricted capacities on the allowed source-target edges. A cut shows that feasibility is equivalent to every target set receiving enough supply from its adjacent sources. For S, excluded supply is $p/a + qd/(ab)$, so its demand minus available supply, multiplied by ab, is

$$sa - a\beta + pb + qd = (s + p - \beta)a + d(p + q).$$

Every source is adjacent to x, so a deficient set cannot contain x. Source y sees every target and supplies at least one target's demand, excluding singletons. If $s \ge k+1$, any excluded source and x would have at least $k + 1$ common neighbors in H. Thus no source is excluded and the set cannot be deficient. For $s \ge 2$, choose two members of S. Their common H-neighbors include x and every excluded source, proving $p + q \le k - 1$. Source y is not excluded, giving $p + 1 \le \alpha$. Finally two excluded exclusive sources have $S \cup \{y\}$ among their common H-neighbors, so $s+1 \le k$ if $p \ge 2$. □

For $k = 1$ no witness is possible, giving the special case in Corollary 1.3. The same elementary witness bounds give an additional useful scalar filter.

Proposition 2.2 (Universal scalar bound and a small side)**.** *Put $b_k = \max\{k + 1, 2k - 2\}$. Every deficient set satisfies $s + p \le b_k$ and*

$$0 < F \le U := (b_k - \beta)a + d(k - 1).$$

Consequently $U \le 0$ certifies triangle equality. If $U > 0$, then

$$\min(a, \beta) < 2b_k.$$

In particular, when $n \le k^2 + k + 1$, every surviving candidate satisfies $a\beta = \mathcal{O}(k^3)$.

Proof. If $p \le 1$, Lemma 2.1 gives $s + p \le k + 1$; otherwise it gives $s + p \le 2k - 2$. Substitute this and $p + q \le k - 1$ in (5). Equality is equivalent to the absence of a positive Hall deficiency, so $U \le 0$ is a valid certificate.

If $\beta < 2b_k$, the small-side conclusion is immediate. Otherwise $U > 0$ gives

$$(\beta - b_k)a < d(k - 1) \le \beta(k - 1),$$

whence $a < 2(k-1) \le 2b_k$. Finally $a+\beta \le n = \mathcal{O}(k^2)$ at small order, and the product of a quantity of order k with one of order k^2 is $\mathcal{O}(k^3)$. □

We also record a general localization bound, useful for rejecting candidates before constructing any rows.

Proposition 2.3 (General localization)**.** *For $k \geq 2$, failure of triangle equality implies*

$$\beta \leq k^2 - k + 1, \qquad d_H(x), d_H(y) \leq k^2.$$

Proof. Let S be deficient. Its excluded mass is at most $(k-1)/a$, while source y alone gives $b/a < s$. Therefore

$$\beta < s + (k-1)b/a < ks.$$

If $s \leq k-1$, this is stronger than required. If $s = k$, Lemma 2.1 gives $p \leq 1$. The case $\beta < k + p$ is immediate. Otherwise put $u = \alpha - p \geq 1$ and $e = p + q \leq k - 1$. As $a \geq \alpha + q$ and $k + p - \beta \leq 0$,

$$\begin{aligned} F &\leq (k + p - \beta)(\alpha + q) + (\beta - \alpha)(p + q) \\ &= ek - u(\beta + q - k) \leq (e + 1)k - \beta - q. \end{aligned}$$

The last step uses $\beta + q - k \geq 0$. Moreover

$$k^2 - k + 2 - \big((e + 1)k - q\big) = (k - 1 - p - q)(k - 1) + (1 - p) \geq 0.$$

Thus $0 < F \leq k^2 - k + 2 - \beta$, and integrality proves the result. The endpoint degree bounds follow from (2). □

3 A sharp finite-order cutoff

For a Hall set write $L = s + p - \beta$. Under padding by isolated vertices of H, the quantities $s, p, q, \alpha, \beta, d$ stay fixed while a increases. Thus $F = aL + d(p + q)$ explains why witnesses with $L < 0$ are transient.

Theorem 3.1 (Cutoff and persistent target bound)**.** *Every deficient Hall set with $L < 0$ satisfies*

$$F \leq k^2 + k + 2 - n, \qquad n \leq k^2 + k + 1.$$

For $k = 2$ no such set exists. Consequently, for $n \geq k^2 + k + 2$, any edge failing triangle equality satisfies

$$\beta \leq b_k := \max\{k + 1, 2k - 2\}. \tag{6}$$

For $k \geq 3$, the order cutoff, the bound $\beta \leq 2k - 2$, and the common endpoint bound $d_H(x), d_H(y) \leq 3k - 3$ are sharp, with the endpoint bound attained at x.

Proof. Put $e = p + q$, $u = a - e$, and $h = \beta - s - p$. By (4), $u \geq 1$, and $L < 0$ gives $h \geq 1$. Since $d = \beta - \alpha$, rewriting (5) gives

$$F = e(s + p - \alpha) - hu > 0.$$

All quantities are integers and $\alpha \geq p + 1$, so

$$hu \leq e(s - 1) - 1.$$

Since $(u - 1)(h - 1) \geq 0$, it follows that $u + h \leq hu + 1 \leq e(s - 1)$. Therefore

$$a + \beta = u + e + s + p + h \leq (e + 1)s + p. \tag{7}$$

For $p \le 1$, the right side is at most k^2+1. For $p \ge 2$, we have $s \le k-1$ and $p \le k-1$, giving at most k^2-1. Adding $r \le k$ proves the order bound. More precisely, with $N = k^2+k+1$, direct expansion gives the slack identity

$$N+1-n-F = \big[N-((e+1)s+p+r)\big] + e(\alpha-p-1) + (u-1)(h-1). \tag{8}$$

All terms on the right are nonnegative. This proves $F \le N+1-n$; for $p \ge 2$ the first term is at least two, giving $F \le N-1-n$. If $k=2$, the earlier inequality $hu \le e(s-1)-1 \le 0$ contradicts $h,u \ge 1$.

Above the cutoff, any deficient set has $L \ge 0$, hence $\beta \le s+p$. If $p \le 1$ this is at most $k+1$; otherwise it is at most $2k-2$. Equation (2) gives the endpoint bounds. Sharpness is established by the following explicit constructions. □

Proposition 3.2 (Numerical boundary types and one-vertex repair)**.** *Let $k \ge 3$, $n = k^2+k+1$, and suppose an edge has a deficient Hall set with $L < 0$. Then*

$$r = s = k, \quad p = 1, \quad q = k-2, \quad \alpha = 2, \quad F = 1, \quad b = k^2-1,$$

and exactly one of the following numerical types holds:

$$(a,\beta,L) = (k, k^2-k+1, -k(k-2)), \qquad (k^2-k-1, k+2, -1).$$

The deficient target set is unique. All residual costs are at most two, $\kappa_0(x,y) = c/b - 1/(ab)$, and adjoining one universal vertex restores triangle equality at the marked edge. This is not a uniqueness assertion about the whole graph.

Proof. At $n = N$, equation (8) and the integer inequality $F \ge 1$ force $F = 1$ and all three nonnegative slack terms to vanish. The branch $p \ge 2$ is impossible. In the other branch, equality in $(e+1)s+p+r \le k^2+k+1$ forces $p=1$, $e = k-1$, $s = r = k$. Since $e > 0$, we get $\alpha = 2$. Also $(u-1)(h-1) = 0$ and $uh = k(k-2)$, so either $u = 1, h = k(k-2)$ or $h = 1, u = k(k-2)$. Substitution gives the two types and $b = k^2-1$.

Write $A = \{y,z\}$. The pair x,z already has all k vertices of S as common H-neighbors. Hence z sees every vertex of R and every target outside S in G. For $t \in S$, the pair y,t has z as a common H-neighbor, and cannot also have all k vertices of R. Some $r_t \in R$ therefore gives a length-two G-path z, r_t, t. Common-source exceptions have distance two through y, and y sees every target. Thus every residual cost is at most two.

Any positive Hall witness has $p \le 1$ and $s \le k$, while $\beta \ge k+2$, so it too has negative slope and numerator one by the same argument. Its excluded exclusive source must be z and its k targets must be exactly $N_H(z) \cap B = S$. This proves witness uniqueness. The maximum cost-one flow leaves exactly one unit, which can be sent at cost two, giving the curvature formula.

Adjoin one universal vertex. For a nonempty fixed target set, the old counts α, β, d, p, q stay fixed, while a increases by one, so its new numerator is $F+L$. Positive old numerators become $1+L \le 0$. Every nonempty target set has $L \le 0$: sets containing x exclude no source; for other sets of size at most k use $p \le 1$ and $\beta \ge k+2$; larger sets exclude no source by codegree. Old nonpositive numerators therefore cannot increase. For the empty set the added common source is excluded too, and the deficit remains zero. Weighted Hall proves the asserted repair. □

For sharpness of the order cutoff, adapt the four-block template of Chen–Liu–You, Section 6 [9], as follows. Take x, y, z, disjoint sets W, S, T of sizes $k-2, k, k(k-2)$, and k further vertices R. In H put exactly

$$x \sim (S \cup T), \quad z \sim (S \cup \{y\}), \quad W \sim S, \quad R \sim \{x,y\}.$$

Here $P \sim Q$ means all edges between the indicated disjoint sets. The order is k^2+k+1. At xy in G,

$$a = k, \quad b = k^2 - 1, \quad \alpha = 2, \quad \beta = k^2 - k + 1, \quad c = k - 2.$$

The set S excludes z and W, so $p = 1$, $q = k-2$, $L = 2k - k^2 < 0$, and $F = 1$ for $k \geq 3$. To check (1), put $D = \{x, z\} \cup W$. Every vertex of S has neighborhood D, of size k; pairs in D share at most S. The pair x, y shares exactly R, also of size k. A pair y, s for $s \in S$ shares only z; pairs involving R have at most two common neighbors, and the remaining pair types have at most one or are already covered. The complement is connected for $k \geq 3$, as T connects to both y and z in G, W connects to both x and y, and R connects to S.

The second boundary type is realized by the same complement edge pattern, with $|T| = 1$ and $k(k-2) - 1$ additional isolated H-vertices. More generally, taking $|T| = h$ and no padding, for $1 \leq h \leq k(k-2)$, realizes equality $F = k^2+k+2-n$ at every integer order $3k+2 \leq n \leq k^2+k+1$. The missing residual pairs have distance two through the nonempty set T, and the maximum Hall numerator occurs at S. Adding an H-edge inside T in the first boundary type, or inside the isolated padding set in the second, preserves the boundary parameters and gives different graph edge counts. Proposition 3.2 does not classify all extremal graphs up to isomorphism.

For persistent sharpness, use an exact member of Chen–Liu–You's Section 6 family [9]. Take disjoint sets Z, S, T, R of sizes $k-1, k-1, k-2, k$, together with x, y, and put exactly

$$x \sim (S \cup T), \quad Z \sim S, \quad y \sim Z, \quad R \sim \{x, y\}$$

in H. Add arbitrarily many isolated H-vertices. Pairs in S have $\{x\} \cup Z$ as common neighbors, pairs in Z have $S \cup \{y\}$, and x, y share R; all three counts are k. Other pair types have at most $k-1$ common neighbors or at most two. At xy, $\alpha = k$, $\beta = 2k-2$, $d = k-2$, and $d_H(x) = 3k-3$. The set S has $p = k-1$, $q = 0$, $L = 0$, and $F = (k-2)(k-1) > 0$ for $k \geq 3$. Padding preserves these facts and makes the order arbitrarily large.

4 Indexed construction and total missing incidence

Build dense indexed adjacency for H. For each vertex v, enumerate pairs in $N_H(v)$ and append v to their witness lists

$$\mathcal{L}(u, w) = N_H(u) \cap N_H(w).$$

Reject as soon as a list receives a $(k+1)$st witness. Before rejection, and throughout an accepted run, there are at most $k\binom{n}{2}+1$ updates. Including adjacency construction and connectivity checking, preprocessing costs $\mathcal{O}(kn^2)$ time and space.

For every edge, degrees and the table entry $r = |\mathcal{L}(x, y)|$ determine

$$c = n - 2 - d_H(x) - d_H(y) + r, \qquad \alpha = 1 + d_H(y) - r, \qquad \beta = 1 + d_H(x) - r.$$

Propositions 2.2 and 2.3, together with Theorem 3.1, give constant-time tests for returning c/b. All remaining edges are called candidates; some candidates still attain equality.

For each residual source u let

$$Q = B \setminus \{x\}, \qquad M(u) = \{t \in Q : ut \in E(H)\}.$$

Since $Q \subseteq N_H(x)$ and $u \neq x$,

$$|M(u)| \leq |\mathcal{L}(x, u)| \leq k. \tag{9}$$

This mask bound is the same missing-neighbor principle as Chen, Liu, and You, Claim 5.2 [9]. Every entry outside $M(u)$ has distance one, including target x. For a common source, entries in its mask have distance two through y. For an exclusive source and a missing target t, the number of common G-neighbors is

$$n - d_H(u) - d_H(t) + |\mathcal{L}(u,t)|. \tag{10}$$

If this is positive the distance is two; otherwise it is three via u, x, y, t. Formula (10) is for a nonedge of G; the formula for an edge has an additional -2.

Scale residual masses by ab. Each exclusive source supplies b, each common source supplies d, each target demands a, and total scaled mass is $M = a\beta$. Rows with identical costs may be merged by adding supplies. Any original plan projects by summing rows, and any quotient plan can be split proportionally among the original row supplies. This proves exactness, including unequal supplies. If $d = 0$, common sources disappear.

Lemma 4.1 (Rows and incidences)**.** *Let m denote the retained row count and I the sum of their missing-mask cardinalities. The reduction can be constructed with*

	m	β	I
$n \geq k^2 + k + 2$	$\mathcal{O}(k^3)$	$\mathcal{O}(k)$	$\mathcal{O}(k^3)$
$n \leq k^2 + k + 1$	$\mathcal{O}(k^2)$	$\mathcal{O}(k^2)$	$\mathcal{O}(k^3)$.

After preprocessing, its construction requires at most $\mathcal{O}(k^4 \log(k+1))$ operations per candidate.

Proof. For small order retain all residual sources. There are at most $a \leq n = \mathcal{O}(k^2)$, and (9) bounds I by km. On candidates surviving Proposition 2.2, we also have $m\beta \leq a\beta = \mathcal{O}(k^3)$ and $\min(m, \beta) = \mathcal{O}(k)$.

For large order, $\alpha \leq \beta = \mathcal{O}(k)$. Retain exclusive sources separately and aggregate common sources by mask. With $q_0 = |Q|$, the common masks of size at least two satisfy

$$\sum_{u \in C} \binom{|M(u)|}{2} \leq (k-1)\binom{q_0}{2}. \tag{11}$$

Indeed every target pair already has x as a common H-neighbor, leaving at most $k-1$ common-source witnesses. This is the endpoint-reservation counting device already used in Section 5 of [9]; here it is summed to bound and construct the full transport representation. There are at most $1 + q_0 + (k-1)\binom{q_0}{2} = \mathcal{O}(k^3)$ distinct common masks. Counting each distinct mask once, its total incidence is at most

$$q_0 + 2(k-1)\binom{q_0}{2} = \mathcal{O}(k^3),$$

because $t \leq 2\binom{t}{2}$ for $t \geq 2$. Exclusive rows add at most $\alpha k = \mathcal{O}(k^2)$ incidences.

It remains to construct this representation without scanning C. The endpoint H-degrees are $\mathcal{O}(k)$ on large candidates, so A, B, R are obtained from their lists. Enumerate common sources missing at least two targets from the lists $\mathcal{L}(s,t)$ for pairs in Q, removing vertices in $A \cup B \cup R$. Their identities number $\mathcal{O}(k^3)$. Obtain each mask from $\mathcal{L}(x,u)$ and the indexed membership test for B. For a target t, the number of common sources missing it is

$$d_H(t) - |N_H(t) \cap (A \cup B \cup R)|.$$

Subtract its already enumerated multiple-miss sources to obtain the singleton-mask count. The remaining c common sources have empty mask. Sort and aggregate the masks, and evaluate exceptional exclusive costs with (10). The stated conservative bound covers these operations and deterministic sorting. Small-order construction may scan all vertices and is no larger. □

5 Direct primal transport and the complexity bound

The solver uses a complete transport matrix with costs in $\{1, 2, 3\}$, represented by default cost one and I exceptional entries. No graph-specific bound on optimal dual labels is needed for this part. We give the details to separate the structural result from the classical optimization it invokes.

Write U_i for the retained row supplies and $V_j = a$ for the target demands. Add terminals σ, τ. Put capacity U_i and cost zero on $\sigma \to i$, capacity V_j and cost zero on $j \to \tau$, and capacity $K = M + 1$ and cost c_{ij} on each row-to-target arc. The internal bound K is nonbinding for any flow of value at most M. A full flow has cost D precisely when the corresponding residual transport does, and

$$\kappa_0(x, y) = 1 - \frac{D}{ab} = \frac{ab - D}{ab}. \tag{12}$$

On the small-order branch, all $m\beta$ row-to-target arcs are explicit. On the large-order branch, build one binary segment tree whose leaves are the ordered targets, with cost-zero, capacity-K arcs directed from each helper to its children. A row's cost-one targets are the complement of its sorted mask, hence at most $|M_i|+1$ intervals. For their canonical segment covers add cost-one, capacity-K arcs from the row to tree nodes. Keep each exceptional cost-two or cost-three arc direct.

Lemma 5.1 (Exact segment representation)**.** *The segment network has the same minimum transport cost as the dense network and has*

$$V = \mathcal{O}(m + \beta), \qquad E = \mathcal{O}\big((m + I + \beta)\log(\beta + 1)\big).$$

Proof. Any dense transfer at cost one can be routed through a cover node and down the tree to its target; exceptional transfers retain their direct arcs. No internal capacity binds since the total flow is $M < K$. Conversely the original forward segment network is acyclic between its row and target layers. Decompose any feasible full flow into terminal-to-terminal paths. Each path chooses one row, then a direct exceptional target or a descendant leaf of a cost-one cover node. It therefore projects to a valid matrix transfer at the same cost. There are $\mathcal{O}(\beta)$ tree nodes and tree arcs, $\mathcal{O}((m + I)\log(\beta + 1))$ cover arcs, and I direct exceptional arcs. □

Lemma 5.2 (At most three marginal-cost stages)**.** *Either representation can be solved by at most three shortest-path computations followed by maximum flows in the zero-reduced-cost residual subgraph. Integral feasible potentials may be maintained by a clipped distance update.*

Proof. Start with zero flow and vertex potentials $\pi = 0$, and use reduced costs

$$\widehat{c}(u, v) = c(u, v) + \pi(u) - \pi(v).$$

All initial residual costs are nonnegative. At a stage, compute shortest reduced distances $\ell(v)$ from σ and put $\Delta = \ell(\tau)$. A residual path to τ exists until the required mass is sent. Set

$$\pi'(v) = \pi(v) + \min\{\ell(v), \Delta\},$$

interpreting an unreachable distance as $+\infty$. For every residual arc of reduced cost $z \geq 0$, the shortest-path inequality and truncation imply

$$\min\{\ell(v), \Delta\} \leq \min\{\ell(u), \Delta\} + z.$$

Thus all new reduced costs remain nonnegative, including on arcs incident with unreachable vertices. Augment a maximum flow in the subgraph of residual arcs of new reduced cost zero. Every

augmentation has original marginal cost $\pi'(\tau)-\pi'(\sigma)$ by telescoping, and new reverse arcs also have reduced cost zero. Exhausting this admissible subgraph makes the next shortest reduced distance strictly positive. Integer costs and potentials therefore give strictly increasing integer marginal costs.

While the total flow is less than M, some source supply and some target demand remain unsaturated. Their complete-matrix transfer gives a forward residual σ–τ path of original cost at most three. In the segment representation the corresponding forward path also exists: every internal arc retains positive residual capacity because its capacity is $M+1$. Hence the minimum original marginal path cost is at most three. The first marginal cost is at least one; all subsequent ones increase strictly. There are consequently at most three stages.

At termination, feasible potentials certify optimality. Each original internal arc retains positive forward residual capacity, and every such arc with positive flow has a residual reverse arc. Its reduced cost is therefore zero. Summing this equality over the flow, including a telescoping sum along tree paths, gives

$$D = \sum_i U_i(-\pi(i)) - \sum_j V_j(-\pi(j)).$$

The same potentials satisfy $-\pi(i)+\pi(j) \le c_{ij}$ for every matrix entry. This is a feasible transport dual of value D, proving exact optimality by weak duality. All arithmetic remains integral. □

This is a bounded-cost instance of classical successive shortest-path optimization with maximal augmentation on admissible arcs. Neither the use of feasible potentials nor the three-value cost observation is presented as a new general min-cost-flow algorithm.

Lemma 5.3 (Short residual paths and Dinic)**.** *In the dense network every simple residual path has length $\mathcal{O}(\min(m,\beta)+1)$. In the segment network it has length $\mathcal{O}(\beta)$. If h is either bound, current-arc Dinic computes an admissible maximum flow in $\mathcal{O}(h^2E)$ operations, independently of capacities.*

Proof. In the dense case, deleting both terminals and the smaller row or target side leaves isolated vertices. A simple path visits each deleted vertex at most once and at most one remaining vertex between successive visits. In the segment case, delete the terminals, all targets, and all helpers, a set of $\mathcal{O}(\beta)$ vertices; the remaining source vertices are isolated. Residual reverse arcs preserve these underlying incidences. The same bounds hold in every admissible subgraph.

Dinic increases shortest augmenting-path length between successful blocking-flow phases, so there are $\mathcal{O}(h)$ phases. Within one phase, current-arc pointers discard each failed level arc at most once. Every successful path augmentation saturates a level arc, and a saturated level arc cannot reappear within that phase, since reverse arcs decrease the fixed level. There are $\mathcal{O}(E)$ augmentations with $\mathcal{O}(h)$ path work each. Thus a phase costs $\mathcal{O}(hE)$ and the full maximum flow costs $\mathcal{O}(h^2E)$. This is a standard short-path argument; unbalanced flow algorithms have long exploited the smaller side [14]. □

Proof of Corollary 1.3. Apply the structural equality shortcuts before constructing rows. At small order, Proposition 2.2 gives $\min(m,\beta)=\mathcal{O}(k)$ and $m\beta=\mathcal{O}(k^3)$. The dense network therefore has $E=\mathcal{O}(k^3)$, and each of the at most three admissible maximum flows costs $\mathcal{O}(k^5)$.

At large order, Lemmas 4.1 and 5.1 give $\beta=\mathcal{O}(k)$ and $E=\mathcal{O}(k^3\log(k+1))$. Thus each admissible maximum flow costs $\mathcal{O}(k^5\log(k+1))$. Binary-heap shortest-path searches cost $\mathcal{O}((V+E)\log V)$ per stage and are smaller than these bounds. The indexed construction cost in Lemma 4.1 is also smaller. Multiply by at most $\binom{n}{2}$ candidates and add $\mathcal{O}(kn^2)$ preprocessing.

All capacities and flow values are $\mathcal{O}(n^2)$, the transport cost is at most $3M$, and the clipped potentials stay between zero and three. Arithmetic therefore uses $\mathcal{O}(\log n)$-bit words. Witness tables require $\mathcal{O}(kn^2)$ words. Holding one network at a time suffices: the dense representation has $\mathcal{O}(n^2)$ arcs, and the segment representation has $m, \beta \le n$, $I \le km$, and $\mathcal{O}(nk \log(n+1) + n) = \mathcal{O}(kn^2)$ arcs. Equation (12) gives exact unreduced output. Complete graphs lie in the class and have $\Theta(n^2)$ edges, proving the explicit-output lower bound. □

This direct-network baseline uses ordinary Dinic. The final hybrid in Section 7 replaces large-order networks by matching and audits the classical AOST small-vertex-cover bound for a theoretical small-order substitution. The AOST backend remains unimplemented; neither its bound nor ordinary Dinic is a new general flow method.

Remark 5.4 (Hall deficiency need not equal the cost gap). The equality test uses only cost-one feasibility, but computing a defective edge can require cost-three transfers. For example, let $S = \{s_1, s_2\}$, let C have arbitrary size c, and put exactly $u \sim (\{y\} \cup S \cup C)$ and $x \sim S$ in H. Its maximum pair codegree is two. At xy in G, $A = \{y, u\}$, $B = \{x, s_1, s_2\}$, $a = c + 2$, $b = c + 3$, and $d = 1$. Vertex u is pendant at x in G. Its supply b exceeds target x's capacity a by one, which must travel distance three to S. The remaining mass can use cost-one arcs. The maximum scaled Hall deficiency is one, but the extra transport cost is two. This persists at arbitrarily large order.

6 A canonical witness when all costs are at most two

The cubic row count need not be paid on every instance. In the one/two-cost regime, an ordinary matching can identify a canonical Hall witness before any common-source masks are materialized. We state the reduction separately from the general algorithm.

For $S \subseteq B$, let $N_A(S)$ be its neighbors among exclusive sources, and define

$$L(S) = |S| - |N_A(S)| - d = |S| + p(S) - \beta.$$

Let ν be the maximum cardinality of a matching in the bipartite graph of cost-one pairs between A and B, and put $\ell = \alpha - \nu \ge 0$.

Proposition 6.1 (Canonical Hall witness)**.** *Assume $a > d(k-1)$. There is a unique inclusion-minimal set $S_0 \subseteq B$ maximizing L. Writing $p_0 = p(S_0)$ and $q_0 = q(S_0)$, the maximum scaled Hall deficiency is*

$$\delta = \max_{S \subseteq B} F(S) = a\ell + d(p_0 + q_0).$$

If every residual cost is at most two, then $D = M + \delta$ and $\kappa_0 = c/b - \delta/(ab)$.

Proof. The matching deficiency formula gives $\max_S(|S| - |N_A(S)|) = \beta - \nu$, hence $\max L = \ell$. The neighborhood-cardinality function is submodular, so L is supermodular. Its maximizing sets are closed under union and intersection; their intersection is the unique inclusion-minimal maximizer S_0.

For any other maximizing set S, we have $S_0 \subseteq S$. The excluded-source counts are antitone under inclusion, so $p(S) + q(S) \le p_0 + q_0$. Thus $F(S) \le F(S_0)$ at the maximal slope. If a positive-deficiency set has smaller slope, integrality gives $L(S) \le \ell - 1$, and Lemma 2.1 gives

$$F(S) \le a(\ell - 1) + d(k-1) < a\ell \le F(S_0).$$

Nonpositive deficiencies cannot exceed $F(S_0) \ge 0$. This proves the formula. For costs one and two, maximum cost-one flow leaves exactly δ units to send at cost two, proving the remaining claims. □

The set S_0 is obtained by alternating reachability from unmatched vertices of B after a maximum matching, or equivalently as the smallest source side of a minimum cut in the usual matching network. This follows because the cut determined by S has capacity $\beta - |S| + |N_A(S)|$; residual reachability gives the unique smallest minimizing source side.

Above the cutoff, an edge with $d \geq k$ attains triangle equality: a deficient set would have $L \leq s - d - 1 \leq k - d - 1 < 0$, contrary to Theorem 3.1. On the remaining candidates, $d \leq k - 1$, $\beta \leq b_k$, and

$$a = n - \beta - r \geq k^2 + 2 - b_k > (k-1)^2 \geq d(k-1).$$

The arithmetic hypothesis of Proposition 6.1 is therefore automatic.

All costs are at most two precisely when no exclusive missing-target pair has distance three; this can be checked in $\mathcal{O}(k^2)$ operations from the indexed pair lists. Constructing the exclusive bipartite graph costs $\mathcal{O}(k^2)$ and ordinary augmenting-path matching costs $\mathcal{O}(k^3)$. If $d = 0$, the formula is simply $\delta = a\ell$ and no common-source query is needed. If $S_0 = B$, then $p_0 = q_0 = 0$ and $\ell = 0$, so the result is equality. For $d > 0$, any other S_0 excludes x and has at least two elements: $L(\varnothing) = -d < 0$ and singletons have $L \leq -d < 0$, whereas $\ell \geq 0$. Choose distinct $s, t \in S_0$. Every common source excluded by S_0 belongs to $\mathcal{L}(s,t)$, which has at most k entries. Filtering that one list for membership in C and adjacency in H to all of S_0 computes q_0 in $\mathcal{O}(k^2)$ operations, without scanning C.

Thus this specialization takes $\mathcal{O}(k^3)$ operations per surviving large-order edge and bypasses the cubic collection of common-source rows, including on the family below. The matching and minimum-cut facts are classical; the candidate additional observation is that the localized Hall arithmetic makes one canonical slope maximizer sufficient. Using the Hall deficit alone fails on the distance-three example above. Section 7 supplies the exact star correction and extends matching to every large-order candidate. The optional canonical-witness implementation agrees with full exact transport on 137,414 eligible edges in the exhaustive six-vertex suite; 81,080 ineligible edges are skipped. Twelve large-order cubic families for $k = 5, \ldots, 16$ also pass without materializing any common-source rows. These cases overlap the general verification suite and are not additional distinct graph coverage. On the actual cubic graph with $k = 20$ and $n = 857$, three-run median per-edge calls took 57.049 milliseconds for the general reducer and 0.153 milliseconds for the canonical specialization. Both exclude the same 170.359-millisecond preprocessing step. This is one structured family on an uncontrolled host, not a whole-graph or application-level speedup.

7 Distance-three stars and complete large-order matching

The one/two-cost restriction can be removed at large order. This uses another elementary complement count and reuses the one canonical matching. Related two-set distance counting occurs in Chen–Liu–You, Claim 5.4 [9]; neither covering counts nor matching are claimed as new general methods. Li–Lu also exhibit a weighted distance-three row bottleneck in their Figure 1 example for LLY transport [3]; the reduction here concerns the localized zero-idleness problem.

Lemma 7.1 (Four-codegree cover)**.** *If G has two distance-three pairs with four distinct endpoints, then $n \leq 4k$. Consequently, when $n > 4k$, the bipartite support of residual cost-three entries is empty or a star.*

Proof. If $\operatorname{dist}_G(u,t) = 3$, then $N_H(u) \cup N_H(t) = V$: every other vertex fails to be a common G-neighbor, and the endpoints are included because $ut \in E(H)$. For a disjoint pair (v,w), intersect the two covering unions. Their four intersections $N_H(u) \cap N_H(v)$, $N_H(u) \cap N_H(w)$, $N_H(t) \cap N_H(v)$, $N_H(t) \cap N_H(w)$ each have at most k vertices and cover V.

Thus no two residual cost-three pairs can be disjoint above $4k$. In a bipartite graph this forces all nonempty support to share one row or one column: fix an edge; edges through its two different endpoints with different other endpoints would be disjoint. The bipartite condition is used here; a general graph of matching number one can also be a triangle. □

We record a bounded-label dual identity to justify the exact star correction. Let w denote scaled source supply, $E(S)$ the sources with no cost-one pair into S, and $Q = B \setminus \{x\}$. Then

$$F(S) = a|S| - M + w(E(S)), \qquad \delta = \max_{S \subseteq B} F(S), \qquad F(\varnothing) = 0.$$

Every singleton has nonpositive deficiency because source y supplies $b \geq a$ and sees every target. Sets containing x also have nonpositive deficiency. For $\varnothing \neq T \subseteq S \subseteq Q$, let $P(S,T)$ be the total supply of sources in $E(S)$ having a cost-two pair into T.

Lemma 7.2 (Nested Hall identity)**.** *The exact transport excess is*

$$D - M = \max\left\{\delta,\ \max_{\varnothing \neq T \subseteq S \subseteq Q} \big(F(S) + F(T) - P(S,T)\big)\right\}.$$

If $Q = \varnothing$, the inner maximum is omitted, leaving δ.

Proof. Use the integral dual labels from Lemma A.1, with target x fixed at zero. Write S for targets of label at most -1 and T for those of label -2. When T is nonempty, the best feasible source label is

$$-1 + \mathbf{1}_{u \in E(T)} + \mathbf{1}_{u \in E(S),\ \text{all costs from } u \text{ to } T \text{ are three}}.$$

This follows by taking the minimum of target-label-plus-cost and the anchor bound one. The source contribution is $-M + w(E(T)) + w(E(S)) - P(S,T)$ and the negative target contribution is $a(|S| + |T|)$. Subtracting M gives the displayed nested expression. When T is empty the expression is $F(S)$. Every nested pair defines feasible labels, and the bounded-label optimum is included, proving equality. □

Proposition 7.3 (Exact star formulas)**.** *If all cost-three pairs share one target, then $D - M = \delta$. If they share one exclusive source z, put*

$$Q_z = \{t \in B : c_{zt} > 1\}, \quad Z = \{t \in B : c_{zt} = 3\}, \quad \delta_Z = \max\big(0, \max_{T \subseteq Z} F(T)\big).$$

Then

$$D - M = \max\big\{\delta,\ b - a(\beta - |Q_z|) + \delta_Z\big\}.$$

Proof. For a column star at t, if T contains another target, every source in $E(S)$ has a cost-two pair into T. The nested expression is then $a|S| - M + F(T) \leq \delta$. If $T = \{t\}$, its singleton deficiency is nonpositive, so the expression is at most $F(S) \leq \delta$. This uses the graph residual's singleton condition; it is false for arbitrary supplies and costs.

For a row star, the same argument applies unless $z \in E(S)$ and $T \subseteq Z$. In that case $P = w(E(S)) - b$, and the expression is $a|S| - M + b + F(T)$. Here $S \subseteq Q_z$, and taking $S = Q_z$ maximizes it for a fixed nonempty $T \subseteq Z$. Allowing the empty choice in δ_Z is harmless only after the outer maximum: $b - a(\beta - |Q_z|) \leq F(Q_z) \leq \delta$. This proves the formula. □

Corollary 7.4 (Complete codegree-two defect formula)**.** *Assume the complement has pair codegree at most two, at any graph order. Triangle equality fails at an oriented edge xy precisely when*

$$A = \{y, z\}, \qquad B = \{x, s, t\}, \qquad zs, zt \notin E(G).$$

In that case $b = a + 1$, the unique positive Hall witness is $\{s, t\}$ with numerator one, and

$$\kappa_0(x, y) = \frac{c}{b} - \frac{1 + \mathbf{1}_{\deg_G(z)=1}}{ab}.$$

Every other edge attains c/b. This is an explicit special-case evaluation of the preceding Hall and star formulas, not a new general transport principle. Priority for the classification remains open; the nearby LLY row-bottleneck precedent in Li–Lu [3] concerns a different measure convention.

Proof. For a positive witness S, $|S| = 2$, the anchor x is excluded, and $p+q \le 1$. The codegree-two exception in Theorem 3.1 rules out negative slope, so

$$3 \le \beta \le |S| + p \le 3.$$

Thus $\beta = 3$, $p = 1$, $q = 0$, and $F = d > 0$. Since $\alpha \ge p+1 = 2$ and $d = \beta - \alpha$, we obtain $\alpha = 2, d = 1$. This gives the stated pattern and its unique target set. Conversely that pattern excludes source z from $\{s, t\}$; codegree two leaves no excluded common source, so $F = 1$ and equality fails.

There is a direct transport proof of the value. Put $\ell = \min\{\mathrm{dist}_G(z, s), \mathrm{dist}_G(z, t)\} \in \{2, 3\}$. Source z supplies $b = a+1$ units and has only cost-one target x, of demand a, so every plan costs at least $M + \ell - 1$. Send a units from z to x and one unit to a nearest target in $\{s, t\}$. Each common source supplies one unit and sees at least one of s, t, since x, z exhaust their common H-neighbors. Send those c units by cost-one arcs. Neither target receives more than $c + 1 < a = c + 2$ from these transfers, so source y can fill the remaining demands by cost-one arcs. This realizes $M + \ell - 1$ without using the nested dual identity.

Finally x, z already exhaust $N_H(s) \cap N_H(t)$. If both distances from z are three, every G-neighbor of z is a common H-neighbor of s, t, hence must be x. Conversely a pendant z at x has distance three to both targets, since x misses them and z, x, y, s and z, x, y, t are paths. Thus $\ell - 1 = 1 + \mathbf{1}_{\deg_G(z)=1}$, proving the formula. A separate exhaustive replay checks the classification and value on 179,074 eligible edges of 21,199 connected labeled graphs through order six, including 622,148 Hall subsets; these overlap earlier suites. The numerical witness type is checked in Lean, while this complete transport construction is not. □

Proposition 7.5 (One canonical matching suffices)**.** *Assume $k \ge 2$ and $a > d(k-1)$, and let S_0 be the inclusion-minimal maximum-slope set in Proposition 6.1, so $\delta = F(S_0)$. In a row-star instance, the exact excess is strictly greater than δ precisely when*

$$d > 0, \qquad Q_z = B \setminus \{x\}, \qquad S_0 \subseteq Z;$$

In that case the excess equals $\delta + d$; otherwise it equals δ. Thus the star correction uses the same one matching that evaluates δ, rather than a second restricted matching.

Proof. The row baseline in Proposition 7.3 is at most d. If $d = 0$, or if z has at least two cost-one targets, it is respectively nonpositive or at most $d - a < 0$, and cannot improve on δ.

Otherwise $Q_z = B \setminus \{x\}$ and $d > 0$. Its deficiency is at least d, since z is excluded. Also $L(Q_z) \ge 0$. If the maximum slope $\ell > 0$, every maximizer excludes x, hence also excludes z; if $\ell = 0$, Q_z itself is a maximizer, so $S_0 \subseteq Q_z$. In either case S_0 excludes z and $\delta \ge a\ell + d$.

If $S_0 \subseteq Z$, the restricted maximum δ_Z equals δ, so the correction adds d. Otherwise no subset of Z has slope ℓ, since every global maximizer contains S_0. If $\delta_Z > 0$, integer-slope dominance gives

$$\delta_Z \le a(\ell - 1) + d(k - 1), \qquad \delta - \delta_Z \ge a - d(k - 2) > d.$$

If $\delta_Z = 0$, use $\delta \ge d$. In both cases $d + \delta_Z$ cannot exceed δ. This proves the criterion, including the empty restricted-set convention. □

Theorem 7.6 (All large-order candidates by matching)**.** *After indexed preprocessing, every edge with $n \ge k^2 + k + 2$ can be computed in $\mathcal{O}(k^3)$ word operations, including distance-three transfers, using matching or a shortcut without constructing common-source rows.*

Proof. Apply the existing equality shortcuts. On remaining large-order candidates, $\alpha, \beta = \mathcal{O}(k)$, $d \le k-1$, and $a > d(k-1)$. For $k \ge 3$, $k^2+k+2 > 4k$, so Lemma 7.1 makes the cost-three support a star. For $k = 2$, a nonstar would need at least two nonanchor source rows and target columns, so $\alpha, \beta \ge 3$. But surviving $\beta \le 3$ and $\alpha \le \beta$ force $\alpha = \beta = 3, d = 0$, giving $U = 0$, a contradiction. Thus there is no exceptional large order requiring flow. Detect all exceptional exclusive pairs and their distances using indexed lists in $\mathcal{O}(k^2)$ operations. One ordinary augmenting-path matching computes δ in $\mathcal{O}(k^3)$, with one $\mathcal{O}(k^2)$ common-source query. Proposition 7.5 tests the row correction in $\mathcal{O}(k)$ and returns D, hence the exact curvature. No common-source row construction is used. □

Proof of Corollary 1.2. At large order use Theorem 7.6. At small order the scalar filter gives $\min(m, \beta) = \mathcal{O}(k)$ and $m\beta = \mathcal{O}(k^3)$. The dense ordinary-Dinic design therefore costs $\mathcal{O}(k^5)$ per candidate, with no segment-tree logarithm. Its construction scans at most $\mathcal{O}(k^2)$ sources, each with at most k exceptions, and sorts their masks in $\mathcal{O}(k^3 \log(k+1)) \subseteq \mathcal{O}(k^4)$ operations. This proves the implemented hybrid bound $\mathcal{O}(k^5 n^2)$.

For the stronger theoretical bound, use the specialized FIFO algorithm of Ahuja, Orlin, Stein and Tarjan, namely the push/bipush maximum-flow algorithm [14], Theorem 5.2 and the vertex-cover extension in Section 8. Its bound is $\mathcal{O}(tE + t^3)$ for a known cover of size t. In each dense small-order admissible network, the smaller row or target side together with the terminals is a cover of size $\mathcal{O}(k)$ and $E = \mathcal{O}(k^3)$, giving $\mathcal{O}(k^4)$ per stage. A fresh source outside the cover with one inlet arc of capacity equal to remaining mass meets the source convention; add zero-capacity reverse arcs as needed. Use a complete feasible maximum augmentation, not a preflow with stranded excess. Removing positive-reduced-cost arcs preserves the cover, and the outer argument still has at most three stages. Shortest paths cost $\mathcal{O}(k^3 \log(k+1)) \subseteq \mathcal{O}(k^4)$ and construction fits the same bound. Large-order matching is already smaller. Add $\mathcal{O}(kn^2)$ preprocessing and at most $\binom{n}{2}$ queries.

Both bounds use the earlier word-arithmetic and unreduced-output convention; preprocessing dominates storage. The AOST substitution is classical and is not implemented or benchmarked here. It gives no claim that the new k exponent is optimal or best known. □

The optional implementation `stage3/star_transport.py` retains the earlier general solver as fallback. It agrees with independent full transport on 164,594 eligible edges in the exhaustive six-vertex suite; 53,900 ineligible edges are deliberately skipped in that optional-backend test. Among the eligible edges, 3,720 row-star cases require a strict correction beyond δ. It also agrees on 2,077 eligible retained GPU-oracle records, skipping 1,911, and on four large pendant-row controls. These protocols overlap earlier coverage. A separately coded subset-based star audit checks 35,072 cost-three cases and the restricted matching formula. Tests corroborate the stated proofs; they do not establish priority. The returned matching/Hall metadata is not a materialized full transport plan.

A separate full-size dual verifier recomputes graph distances by BFS and checks every actual residual source–target inequality and the exact integer objective. It passes 3,734 eligible edges

through five vertices and six larger controls against independent full transport, and rejects three deliberately corrupted certificates. The ordinary certificate uses target labels -1 on S_0 and zero elsewhere, and source label one exactly on $E(S_0)$; shift all labels together when $S_0 = B$ to anchor x at zero. For a strict row correction, the nested pair $S = Q, T = S_0$ supplies the labels in Lemma 7.2. A feasible dual alone proves a lower bound, not a primal plan attaining it. This verifier scans common sources and is excluded from the fast runtime and benchmark clocks; its graph coverage overlaps the earlier suites.

8 Cubic row counts on defective edges

Proposition 8.1. *For every $k \geq 5$ there is a connected graph satisfying (1), of order at least k^2+k+2, with a triangle-defective edge whose residual transport has at least $\binom{k-2}{3}$ distinct positive-supply common-source cost rows. Its exact deficit is $(k-1)/(ab)$.*

Proof. Take x, y, disjoint sets S, T, Z of sizes $2, k-2, k-1$, and one vertex c_U for each three-element subset $U \subseteq T$. In H put exactly

$$x \sim (S \cup T), \qquad y \sim Z, \qquad Z \sim S, \qquad c_U \sim U \quad (U \in \tbinom{T}{3}).$$

Pad by isolated H-vertices if needed. Two vertices of S have the k common neighbors $\{x\} \cup Z$. Two vertices of T have x and the $k-4$ triple vertices containing both, totaling $k-3$. Two vertices of Z share $S \cup \{y\}$, of size three. The pair x, c_U shares U, of size three; distinct triple vertices share at most two targets. The pair y, s, $s \in S$, shares Z, of size $k-1$. All other types have at most two common neighbors. Thus (1) holds.

At xy, $A = \{y\} \cup Z$, $B = \{x\} \cup S \cup T$, $\alpha = k$, $\beta = k+1$, and $d = 1$. Every c_U is a common source with mask exactly U and scaled supply one. Their rows are distinct. The complement is connected: Z connects to x, $S \cup T$ to y, and every common source to both endpoints. Missing common-source transfers have length two through y; missing Z-to-S transfers have length two through a member of T.

The set S has $s = 2$, $p = k-1$, $q = 0$, $L = 0$, and $F = k-1 > 0$. Sets meeting both S and T exclude no sources. A singleton from S has the same excluded sources but a smaller numerator. A subset of T excludes only common sources and has negative L; by Theorem 3.1 it cannot be deficient at this order. Sets containing x are harmless. Thus $k-1$ is the maximum scaled Hall deficiency.

When all residual costs are one or two, maximum flow on cost-one arcs sends precisely total mass minus the maximum Hall deficiency. Completing its unmatched supply and demand at cost two is feasible, and no plan can use less cost-two mass. The extra cost over M is therefore $k-1$, proving the claimed deficit. □

The proposition proves sharp cubic order for the number of distinct rows retained by identical-row aggregation. It is not an $\Omega(k^3)$ lower bound for arbitrary curvature algorithms or every possible compression. In this family the triple rows do not create the actual deficient set S; a more semantic shortcut could bypass them.

9 Verification, timing, and limitations

Verification compares the graph solver with a separately implemented full-neighborhood integer transport reference. The suite includes every connected labeled graph of orders two through six

(27,475 graphs and 218,494 edges), seeded larger cases, and the cubic family for $k = 5, \ldots, 15$, for 27,583 graph instances and 218,956 exact edge comparisons overall. Both the earlier closure backend and a replay with the revised primal backend and final scalar filter pass. The final run records 92,853 shortcut returns. These are repeated checks of overlapping instances, not twice as many sampled graphs. Selected tests force residual construction despite an applicable shortcut. Finite verification corroborates the proofs and detects implementation errors; it does not prove the universal theorem.

The primal backend also passed 21,279 complete ternary cost matrices for the dimensions through 3×3 listed in its script, with seeded integral marginals, both dense and segment representations, and an independent min-cost reference. This is exhaustive over those cost matrices, not every possible marginal vector. At most three marginal stages were observed.

A separate CUDA oracle samples 4,096 marked-edge graphs of order nine with seed 2026102104, retaining 3,988 connected records. It enumerates all 3^8 anchored source-label assignments per retained record and chooses target potentials optimally; Appendix A justifies the labels. Unused source labels duplicate assignments. The small integer-valued float32 operations are exact with TF32 disabled. Every retained value agrees with a forced CPU solve, including a replay with the revised primal backend. This is exhaustive label enumeration for sampled graphs, not exhaustive graph coverage. Its recorded 8.254-second run on an NVIDIA GeForce RTX 5090 has no matched CPU timing; no GPU speedup is claimed.

The initial timing study compared dense and segment versions of a two-level maximum-closure reduction, using identical current-arc Dinic code. It is a historical representation experiment, not a timing of the current primal algorithm. Table 1 retains selected original measurements. Each entry is the median of three runs, including construction and certificates, on the synthetic triple-mask family under Python 3.14.7 and Windows 10 build 19045, with recorded processor string AMD64 Family 25 Model 33 Stepping 2.

k	Rows	Dense arcs	Segment arcs	Dense (s)	Segment (s)
8	22	428	338	0.000988	0.002067
16	366	12,500	6,832	0.033505	0.065597
32	4,062	268,196	96,742	1.353074	1.914443

Table 1: Historical matched closure encodings with identical Dinic solver. The segment encoding is slower at every original tested size. Arcs exclude residual reverse bookkeeping.

At $k = 32$ that segment encoding used about 2.77 times fewer arcs but took 1.41 times as long, with 19 blocking-flow phases versus 3 and longest observed augmenting paths of lengths 21 versus 5. These observations are consistent with helper and path overhead, but do not isolate causality. A separate $k = 16$ ablation gave 0.764486 seconds for dense Edmonds–Karp, 0.028342 for dense Dinic, and 0.054148 for segment Dinic. The roughly 27-fold baseline improvement reflects a classical solver switch and implementation, not a new flow method.

The new comparison in `batched-benchmark.json` runs closure and primal backends on identical synthetic complete transport inputs with default cost one and exceptional cost two or three. The cost-three variant is a synthetic transport matrix, not asserted to be realized by the cubic graph construction. Table 2 reports the latest $k = 32$ medians of three runs, including network construction and certificates. Host load was uncontrolled; these are local exploratory measurements.

For the cost-one/two inputs the corresponding arc counts are 268,196 and 96,742 for the two closure encodings, and 138,141 and 52,414 for the primal encodings. Dense primal flow is faster than segment primal flow in both displayed comparisons despite the larger arc count. The primal

Costs	Closure dense	Closure segment	Primal dense	Primal segment
$\{1, 2\}$	1.339819	2.040211	0.574793	0.625651
$\{1, 3\}$	1.157711	1.316381	0.590050	0.676118

Table 2: Latest shared-host synthetic times in seconds at $k = 32$. The primal backend is classical batched min-cost flow. These measurements do not establish a segment advantage or an application speedup.

backend does not win every comparison in the complete record: at $k = 8$ with cost-one/two inputs, dense closure is faster than dense primal flow. All comparisons concern synthetic inputs and classical backend choices; they do not demonstrate the worst-case parameter bound or a general practical benefit from sparse representation.

A conventional specialization is available when every residual cost is one or two: if F_1 is maximum cost-one flow, then $D = 2M - F_1$. A separate implementation passed 454 actual-graph comparisons across 53 graphs and skipped 180 cases with cost-three entries. Its $k = 32$ segment-versus-dense ordering reversed in a fresh replay, further limiting any crossover claim. This specialization must not be extended to arbitrary cost-three instances.

The reproducibility materials include both solver generations, the indexed reducer, independent full transport, matrix and graph verification scripts, CUDA oracle and replay, and timing records. The previous-stage public review draft is available at https://github.com/Autio/autio.github.io/pull/2; the separately versioned stage-three materials record the current corrections and checks. A draft pull request is not an arXiv submission or independent review. The deterministic theorem does not measure Python allocation, hashing, arbitrary-integer arithmetic, or fraction normalization. Negative timing results remain part of the record.

9.1 Established software and the final matching extension

A new fixed comparison uses ten complete transport matrices and 22 graph workloads on 19 distinct graph instances, with five timed trials after an untimed warmup. Backend order is seeded and shuffled each trial. The process is pinned to one logical CPU and native thread limits are one. The host is a Ryzen 5 5600 with approximately 32 GiB RAM, Windows 10, Python 3.14.7, NumPy 2.5.4, SciPy 1.18.1, NetworkX 3.7 and POT 0.9.7.post1. The host is not isolated: concurrent research work can affect timings. Raw trials, ranges, medians and median absolute deviations (MAD) are retained, with no outlier removal. MAD is descriptive dispersion, not a confidence interval.

Conventional graph baselines already cancel common mass, aggregate identical rows where applicable, and dispatch equal-degree instances to SciPy assignment. Other branches use NetworkX integer network simplex, SciPy HiGHS, or POT's native CPU network simplex [5, 6, 7]. Thus the comparison does not deliberately handicap the baselines by insisting on an unreduced full-neighborhood LP. Timings include per-query representation work and the checks made by each backend. Indexed methods include a fresh global index in every cold batch. Input generation, selection of the valid parameter and query list, imports and process startup are excluded.

HiGHS and POT are floating-point algorithms. In these recorded runs, their rounded integer plans are checked for exact row/column balance and nonnegativity, and rounded duals satisfy every cost inequality with exactly matching integer primal/dual objectives. This certifies the reported optima without describing the libraries as exact-arithmetic solvers. Assignment outputs are checked feasible permutations and agree in integer objective across backends. HiGHS is not run above a declared 20,000-variable budget; this omission is not a measured timeout.

Native POT has the lowest median on all ten generic transport matrices. For example, on the 4062×33 triple-mask cost-one/two matrix, dense and segment Python batched methods take 575.712 ± 9.236 and 424.808 ± 19.544 milliseconds, while POT takes 57.668 ± 0.345 milliseconds. This rejects a general practical transport-speedup interpretation of our Python flow implementation.

The graph results are mixed. The earlier two-cost canonical method takes 30.435 ± 0.249 milliseconds on all 2,612 edges of the $k = 8, n = 74$ cubic graph, versus POT dispatch at 279.302 ± 6.196 milliseconds. Conversely, on only eight queries of the $k = 24, n = 1589$ cubic graph, cold canonical takes 1079.377 ± 22.800 milliseconds versus POT 185.074 ± 0.748 milliseconds; median canonical preprocessing alone is 998.498 milliseconds. On the 128-vertex path, general adaptive indexing takes 171.270 ± 7.060 milliseconds for all 127 edges, versus 1.791 ± 0.009 milliseconds for POT's assignment dispatch. These are workload-specific observations, not general speedup claims or an empirical proof of the asymptotic bound.

A separate final 15-workload supplement evaluates the single-matching star implementation, using the same thread/order protocol and five trials per cold and warm mode. It preserves the original 22-workload record. Two earlier two-matching pilots are documented, but none of their timings is mixed into the final single-matching data. Table 3 gives the two actual all-edge star-family workloads; every indexed cold batch includes preprocessing.

Workload	Queries	Star matching	Old canonical	POT dispatch
Row-star, $n = 55, k = 2$	1430	37.626 ± 9.294	43.142 ± 6.730	60.912 ± 8.309
Column-star, $n = 57, k = 3$	1531	64.265 ± 16.870	88.874 ± 32.402	131.828 ± 31.535

Table 3: Final single-matching supplement: cold milliseconds, median ± MAD. Shared-host variability prevents a precise speedup interpretation. The second workload also contains row-star edges.

The final supplement is particularly noisy: the cold $n = 205$ row-star marked query ranges from 11.893 to 77.472 milliseconds, and some warm batches are slower than their corresponding cold batches. No quieter rerun or outlier filtering replaces these data. On the second all-edge workload, star and POT cold ranges overlap. The stronger new evidence is coverage: all 2,961 queried edges of these two graphs use matching or a shortcut, with no general-flow fallback, while the previous canonical dispatcher makes 100 flow calls. The marked row-star family has scaled Hall deficit one and exact transport excess two, so it genuinely requires the distance-three correction. The marked column-star examples have cost-three pairs but attain triangle equality; those entries need not be used by an optimum.

Warm and cold results must be read together. For the $n = 505$ row-star marked edge, the new warm query takes 0.099 ± 0.006 milliseconds, but its cold query takes 90.954 ± 31.098 milliseconds versus POT 0.815 ± 0.219 milliseconds. Retaining this adverse case is essential. The experiment shows that semantic reduction can remove specific flow calls; it does not establish a robust cold-query advantage, a benefit on sparse empirical graphs, or superiority to production transport software.

10 Research status and AI assistance

This manuscript is a research draft with explicit mathematical arguments for its stated results. A bounded literature search and internal computational checks have been performed. Neither priority nor acceptance by a subject expert has been established, and no independent human mathematical review is recorded. The result should be assessed as a focused candidate contribution, without a claim of a major algorithms breakthrough.

Supporting work contains partial Lean formalizations of finite label constraints, graph neighborhood exclusions, arithmetic localization inequalities, the scalar-filter arithmetic, and the new graph-valued bridge from actual neighborhood sets to the cutoff, small-side and persistent consequences. Separate files check conditional boundary and slack arithmetic. These were checked by the Lean kernel without `sorry` or `native_decide`, with only standard axioms. New files also check weighted Hall by cloning, the actual-graph quantitative numerator bound and codegree-two exception, boundary neighborhood types and two-step paths, and the four-codegree cover. Real residual matrices attain their unit-cost lower bound exactly when the graph Hall inequalities hold, for supplied costs whose unit pairs are graph edges. The canonical least maximizer and weighted priority are checked on finite relations and instantiated from actual graph codegrees. Restricted-set correction arithmetic is checked under an explicit margin; it is not a formal transport identity. Metric common-mass cancellation, general higher-cost transport integrality, complete extremal realization and repair, nested duality, matching program correctness, runtime analysis and the complete manuscript remain outside an end-to-end formal proof. Test success and partial formalization must not be described as a machine-checked proof of Corollary 1.2 or the complete manuscript.

OpenAI Codex agents, including GPT-6 Astra agents, contributed substantially to conjecture exploration, proof development and criticism, literature searches, software construction, test design, interpretation of results, and manuscript drafting. The proposed human attribution on the title page requires the named person's acceptance of authorship and responsibility. This disclosure does not assert that the proposed author has independently checked every proof, citation, or computation. The AI systems are tools and are not listed as authors. The current manuscript and source package remain for human review before any submission or publication decision.

A Finite labels for the independent oracle

This appendix describes a verification device and the earlier closure formulation; the primal algorithm does not depend on it. For supplies U_i, demands V_j, and costs c_{ij}, the transport dual is

$$\max\left\{\sum_i U_i f_i - \sum_j V_j g_j : f_i - g_j \le c_{ij}\right\}. \tag{13}$$

Lemma A.1 (Three labels suffice)**.** *An optimal solution of* (13) *exists with*

$$f_i \in \{-1, 0, 1\}, \qquad g_j \in \{-2, -1, 0\}, \qquad g_x = 0.$$

Proof. Finite metric transport has the Kantorovich dual in graph Lipschitz potentials. Anchor the value at x to zero. The graph edge constraints have a totally unimodular incidence matrix and integral right-hand side, so an integer optimal potential exists; this fact is also proved for graph curvature in [8]. All residual sources are adjacent to x and all targets are within distance two of x through y, giving source values in $\{-1, 0, 1\}$ and target values in $\{-2, -1, 0, 1, 2\}$. Common-mass cancellation leaves its objective unchanged. Identical cost rows can take a common best feasible source value, so aggregation preserves an optimal dual with these bounds.

Replace positive target values by zero. Feasibility is preserved because $f_i \le 1 \le c_{ij}$, and the maximized objective does not decrease. Transport duality implies the resulting feasible solution is still optimal. It need not extend to a graph Lipschitz potential; containment of an optimum in the bipartite feasible family is sufficient. □

Writing each integer label as a sum of two threshold bits turns these inequalities into weighted implications and hence into the classical maximum-closure problem [13]. This was the original experimental backend. Its extra label layer is unnecessary for the direct primal proof above.

B Two exact order-theoretic descriptions

The transport dual and the historical closure encoding admit elementary adjunction descriptions. These clarify its feasible sets but are not additional novelty claims or runtime arguments. We use the monotone adjunction convention; nLab notes that "Galois connection" can also refer to the antitone convention [15, 16].

For a finite cost matrix, order source and target potential vectors coordinatewise and define

$$A(g)_i = \min_j (c_{ij} + g_j), \qquad B(f)_j = \max_i (f_i - c_{ij}).$$

Both maps are monotone, and rearranging the inequalities proves

$$B(f) \le g \quad \Longleftrightarrow \quad f \le A(g).$$

Thus B is left adjoint to A. The composite AB is an extensive idempotent monotone operator on source potentials, while BA is reductive and idempotent on target potentials. Dual feasibility is exactly either side of the displayed equivalence. This tightening structure alone gives neither an optimum for the weighted objective nor an iteration bound.

For the historical implication graph, let $C(S)$ be the vertices reachable from S, including S and any forced anchor. The map is extensive, monotone, and idempotent. Its fixed points are the feasible successor-closed sets containing the anchor. If i includes this fixed-point poset into the powerset poset, then for a closed set T,

$$C(S) \subseteq T \quad \Longleftrightarrow \quad S \subseteq i(T).$$

Hence closure, with codomain $\mathrm{Fix}(C)$, is left adjoint to inclusion. Maximum closure optimizes over this fixed-point family; it is not equivalent to closing an arbitrary initial set. No philosophical interpretation is needed for either statement or for the algorithm.

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